

Application of mobile computing in emergencies

- patient data transmission –
- battlefield communication
- ad-hoc networks during disaster

Mobile computing applications include M-Banking, M-Commerce, and M-Payment.

Mobile computing emphasizes connectivity and access to resources while moving, whereas portable computing refers to devices that can be carried but may not maintain continuous connectivity.

Portable computing is about mobility of the device, while mobile computing is about mobility with continuous network access.

Wireless sensor networks consist of distributed sensors that monitor physical or environmental conditions and transmit data wirelessly, used in fields like healthcare, agriculture, and environmental monitoring.

Wireless sensor networks consist of distributed sensor nodes with limited power, memory, and processing, designed to monitor and transmit environmental data.

WSNs are characterized by small, resource-constrained nodes that collaborate to sense and transmit data.

WSNs provide real-time monitoring and data collection in diverse applications such as healthcare and agriculture.

Characteristics of Wireless Sensor Networks (WSNs)

- Ability to withstand harsh environments/fault-tolerance
- Dynamic network topology
- Energy-efficient operation
- Low bandwidth
- Support large-scale deployment
- Decentralized/self-organizing

Application of WSNs

- forest fire detection,
- military surveillance,
- smart grids,
- civil structural health monitoring, and
- industrial process monitoring.

Scalability is a major challenge in wireless sensor networks due to the large number of nodes and dynamic topology.

Signaling System 7 (SS7) is used in GSM networks for mobility management and roaming support.

Proactive routing protocols maintain fresh lists of destinations by periodically distributing routing tables.

Wireless LANs provide local area connectivity without physical cables, enabling mobility, flexibility, and ease of access to network resources within a limited area.

Wireless LANs are crucial in mobile computing because they allow devices to connect and communicate without wired infrastructure, supporting mobility and convenience.

WiMAX is defined under IEEE 802.16 standards, providing wireless metropolitan area network capabilities providing broadband wireless access up to 30 miles for fixed stations.

IEEE 802.11 defines Wireless LANs (WLANs), commonly known as Wi-Fi.

IEEE 802.11a supports bandwidth up to 54 Mbps and operates at 5 GHz.

802.11b has the slowest maximum speed compared to other Wi-Fi standards.

802.11n uses MIMO technology to improve bandwidth and range.

Middleware for WSNs must handle distributed processing, low bandwidth communication, and real-time computation. Scalability is limited.

Middleware provides services like message queuing and data reduction, allowing applications to continue functioning despite disconnections.

Wireless middleware isolates applications from connectivity issues such as intermittent connections and varying throughput.

Wireless transmission noise includes

- thermal noise,
- impulse noise, and
- crosstalk.

The Base Station Controller (BSC) manages communication between multiple BTSs and the Mobile Switching Center (MSC).

Mobile IP allows a device to maintain ongoing connections while changing its point of attachment, whereas nomadic IP requires a new connection each time the user moves.

Mobile IP was designed to support mobility while maintaining continuous internet connectivity.

Mobile IP uses Agent Advertisement messages to inform mobile nodes of available care-of addresses and whether the agent is home or foreign.

Mobile IPv6 includes built-in route optimization, care-of address transparency, and solves ingress filtering issues. Optional route optimization, limited scalability in IPv4. IPv6 addresses-128 bit, IPv4 addresses-32-bit

The care-of address represents the temporary location of a mobile device while away from its home network.]

Mobile computing environments face

- frequent disconnections,
- heterogeneous networks, and
- device limitations. (small display screens, limited memory)
- Transmission interference

Sky-wave propagation involves signals reflecting off the ionosphere, allowing long-distance communication such as shortwave radio.

Ground-wave propagation follows the contour of the earth and can travel considerable distances.

Fixed hosts perform transaction and data management functions with the help of database servers in mobile database architectures.

MANET /ad hoc (operate without fixed infrastructure, relying on peer-to-peer connections.) routing approaches include

- proactive,
- reactive,
- hybrid, and
- hierarchical methods.

Routing in ad hoc networks involves dynamically determining paths between nodes without fixed infrastructure, often using protocols like AODV or DSR.

Ad hoc routing protocols establish communication paths in networks where topology changes frequently.

Wireless communication uses

- isotropic reflective,
- parabolic reflective, and
- dipole antennas.

Wireless networks transmit data using electromagnetic waves like radio waves, typically at the physical layer of the OSI model.

Wireless networks provide connections where cabling is impossible, often at lower cost, and offer portability and flexibility.

Wi-Fi provides local area connectivity without cables, making it WLAN Wireless Local Area Network technology.

Bluetooth is a wireless personal area network (WPAN) technology used to connect devices within a short range. low-power communication.

PANs enable unconscious communications where devices autonomously detect and connect without user intervention.

Bluetooth is a PAN (personal area network) technology that connects devices within a short range.

Wireless networks include PAN, MAN, and WAN. LAN is typically wired, though WLAN exists as a wireless variant.

Personal Communications Service (PCS) was designed to meet the needs of personalized, anytime, anywhere communications after cellular networks.

Cell splitting involves subdividing congested cells into smaller ones, increasing the reuse of channels and system capacity.

PCS networks in the US use the 1900 MHz frequency band, compared to 800 MHz for cellular networks.

Full duplex systems allow simultaneous two-way communication.

GSM uses TDMA (Time Division Multiple Access) to allow multiple users to share the same frequency channel using non-overlapping time slots.

Frequency reuse allows efficient use of limited spectrum by assigning the same frequencies to different cells, increasing capacity and coverage.

Frequency reuse maximizes spectrum efficiency and supports more users in cellular networks.

GSM technology supports SMS as one of its key features.

CDMA systems use soft handoff where mobiles share the same channels across cells.

Mobile communication bandwidth limitation is orders of magnitude slower than fixed networks.

Mobile networks use

- FDMA,
- TDMA,
- CDMA, and

- SDMA as multiple access methods.

The TCP/IP protocol suite consists of application, transport, and network layers.

2G networks introduced digital communication standards such as GSM and CDMA.

3G networks introduced global roaming and higher data speeds compared to 2G.